



Detailed numerical simulation and experiments of a steadily burning micron-sized aluminum droplet in hot steam-dominated flows

Yue Qiu^{a,*}, Sheng Feng^b, Zhiyong Wu^a, Shijie Xu^{b,c}, Can Ruan^a, Xue-Song Bai^b, Elna J.K. Nilsson^a, Marcus Aldén^a, Zhongshan Li^a

^a Division of Combustion Physics, Lund University, 221 00 Lund, Sweden

^b Division of Fluid Mechanics, Lund University, 221 00 Lund, Sweden

^c School of Mechanical Engineering, Shanghai Jiao Tong University, Shanghai 200240, PR China

ARTICLE INFO

Keywords:

Metal combustion
Micron-sized aluminum droplet
Eulerian-based framework
Direct numerical simulation
Flame structure

ABSTRACT

Detailed numerical simulations are conducted in comparison with experimental results to study the flame structure and burning rate of a steadily burning aluminum droplet in hot steam-dominated environments. The droplet surface temperature, flame temperature, and flame stabilization position are measured along with the droplet burning rate estimated from the droplet size evolution. A numerical model accounting for detailed transport properties and chemical kinetics is presented and applied to unveil the flame structure, species and temperature distributions, and heat/mass transfer between the droplet and the surrounding gas. The numerical results of the temperature, velocity, and species distribution profiles demonstrate that the aluminum vapor flame is of classical diffusion flame structure, where near the droplet, there is a non-negligible amount of AlOAl apart from the main product $\text{Al}_2\text{O}_3(l)$. This supports the deposition and formation of an alumina cap on the surface proposed in the literature. The simulation correctly captured the flame temperature and flame stabilization distance for a range of droplet sizes. Net heat flux analysis shows that conduction heat from the flame front accounts for less than 30% of the heat needed in aluminum evaporation, which warrants further quantification on other heat sources. The experimental and numerical results enrich the knowledge of the heat/mass transfer and chemical reactions near the droplet, which helps deepen the understanding of aluminum droplet burning.

1. Introduction

Inspired by the concept of the metal-enabled cycle for production and transportation of renewable energy [1,2], metal fuels have been a hot topic in recent years, expanding its application from usual energetic fuel additives in solid propellants and additives. Micron-sized aluminum particles can be good candidates as energy carriers because of their rich resources, high energy density, and relatively low cost than nano-sized particles. Burning aluminum in the steam atmosphere, in particular, facilitates the production of heat and hydrogen onboard. More potential massive applications can be further extended to exploration on the Moon.

Optimal use of aluminum as an energy carrier requires an increased understanding of the fundamental combustion properties, including ignition conditions and mechanisms for the formation of combustion products. Experimental and modeling tools are ideally combined to gain an understanding of both macroscopic properties and the mechanisms on a molecular level.

Experimental data on single particle combustion of aluminum particles are, unfortunately, scarce. There have been some studies on their ignition and combustion properties on a macro level back in the 1960s [3–5]. A quantitative spatially-resolved measurement of the aluminum particle combustion in air, and the basis for the majority of the single particle modeling studies, stem from the work by Bucher et al. [6]. The same group also conducted studies in different oxidizers, including steam [7], and summarized the state-of-the-art knowledge in the literature from the year 2000 [8]. The experimental work is on single isolated aluminum particles with a diameter of 210 μm , originating from a laser-heated aluminum wire. The most thorough work is with air as the oxidizer [6] and included characterization of profiles of AlO, temperature, and Al_2O_3 within a distance ten times the radius of the burning particle. Considering the peak AlO concentration as indicative of the flame position, it was seen that the flame is stabilized at a distance corresponding to about three times the particle radius. The temperature was close to alumina's melting

* Corresponding author.

E-mail address: yue.qiu@fysik.lu.se (Y. Qiu).

<https://doi.org/10.1016/j.proci.2024.105717>

Received 4 December 2023; Accepted 22 July 2024

Available online 22 August 2024

1540-7489/© 2024 The Author(s). Published by Elsevier Inc. on behalf of The Combustion Institute. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

temperature at the particle surface, then increasing to a peak value of about 3800 K at five times the radius. It is worth noting that the formation of AlO did not coincide spatially with a significant heat release. Al₂O₃ had its peak slightly further away from the particle than the AlO peak. The data from the case with steam as the oxidizer [7] are limited as a result of experimental challenges, but it is concluded that the flame is closer to the particle surface when burning in steam compared to combustion in air. This same trend is seen in the work of Halter et al. [9], which studied the combustion of particles with diameters of 40–80 μm in an electrodynamic levitator.

Modeling efforts related to single particle combustion of aluminum (Al) up until 2004 were reviewed by Beckstead et al. [10]. More recent work treating the aluminum particle as a point in an Eulerian–Lagrangian framework was performed by Zhang et al. [11]. Glorian et al. [12] and Hu et al. [13] used 2D axisymmetric conservation equations with detailed gas phase and surface chemistry. In both works, evaporation is treated as a surface reaction. The studies mainly focused on air as the oxidizer, but Glorian et al. [12] also included several other oxidizers. The results for steam gave unreasonably low temperatures, which the authors hypothesized could be due to erroneous treatment of H-species in the surface chemistry.

In summary, for aluminum oxidation in steam, the available experimental data mainly indicate trends in comparison with the case of air as an oxidizer; the flame is faint and closer to the droplet surface. The lack of more elaborate experimental data hampers the development of predictive models.

Motivated by the need to characterize the physical and chemical processes occurring close to the burning aluminum droplet, this work presents a new Eulerian-based modeling framework within which important parameters such as temperature, velocity, and key species profiles are resolved near the droplet surface. The model is further validated against experimental data. Interesting phenomena in heat balance, droplet size, and oxidizer dependence are observed and discussed. Results serve to provide some insights into the estimation of mass and heat exchange between the gas phase and droplet phase in the future Lagrangian model.

In the remainder of this paper, the model is first described, with a particular focus on the boundary conditions. This is followed by the presentation of the modeling results and a discussion in relation to experimental data. Conclusions are drawn on the importance of different physical and chemical processes for aluminum burning in steam.

2. Model description

This model considers a detailed simulation of the boundary layer of a burning aluminum droplet in its steady-state combustion phase. Aluminum droplet combustion was observed to consist of two stages with a steady symmetric burning stage followed by an asymmetric burning stage [14]. This paper focuses on the first stage. The droplet temperature is higher than the melting point of the alumina shell (2327 K [15]). Therefore, the alumina shell melts and shrinks to a small cap, and the inner bare aluminum is exposed. The effects of the alumina shell are not considered. The droplet features “thermally thin” assumption, i.e., there is no net heat flux at the boundary. This is substantiated by the Biot number at a high temperature (~2500 K) being $Bi = hL/\lambda_{Al(l)} = Nu \cdot \lambda_{Al(g)}/\lambda_{Al(l)} \approx 0.0014$, whereas Nu is taken as 2, $\lambda_{Al(g)}$ as 0.07 W/mK [12], and $\lambda_{Al(l)}$ as 100 W/mK [16]. Besides, only the gas phase region is modeled. The heat and mass exchange between the droplet and the surroundings is achieved through the droplet surface, which is considered as the *Fuel* boundary in the model. On the boundary, only the evaporation of the aluminum droplet is assumed without considering the potential heterogeneous surface reaction or deposition discussed in Ref. [12,17]. The governing equations for the gas phase and boundary are given below.

2.1. Gas phase equations

The governing equations for mass, species, momentum, and energy in the gas phase are given below.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0, \quad (1)$$

$$\frac{\partial \rho Y_i}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_i) - \nabla \cdot (\rho D_i \nabla Y_i) = \dot{\omega}_{g,i}, \quad (2)$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) - \nabla \cdot \boldsymbol{\tau} + \nabla p = 0, \quad (3)$$

$$\frac{\partial \rho h_s}{\partial t} + \nabla \cdot (\rho \mathbf{u} h_s) - \nabla \cdot (\rho \alpha \nabla h_s) = \rho \dot{Q}_g, \quad (4)$$

where the subscript “g” stands for “gas”; ρ is the gas density; $\mathbf{u}$ is the gas velocity vector; Y_i is the mass fraction of species i in the gas mixture; D_i is the diffusion coefficient of species i in the gas; p is the gas pressure; $\boldsymbol{\tau}$ is the viscous stress tensor of gas; h_s is the sensible enthalpy of the gas mixture; α is the thermal diffusion coefficient. $\dot{\omega}_{g,i}$ and $\dot{Q}_g$ depict the chemical reaction rate for species i and heat release rate due to gas phase reactions, respectively.

2.2. Boundary conditions

The boundary is considered as the droplet surface separating the gas phase and the droplet condensed phase. At the boundary, evaporation transports mass from the droplet to the gas phase. The deposition of gaseous species on the droplet surface is not fully understood and not considered in the current model. Assuming aluminum is the only species coming out from the droplet, denoted as F , the mass flow rate and gas-phase species conservation equations are given as below [12, 18]:

$$\dot{\omega} = \sum_i \dot{\omega}_i, \quad (5)$$

$$\dot{\omega}_i = \dot{\omega}(Y_i)_s - (\rho D_i \nabla Y_i \cdot \mathbf{n})_s. \quad (6)$$

Also,

$$\dot{\omega}_i = \begin{cases} \dot{\omega}_F, & i = \text{Al} \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

where s denotes the interface; i is the i_{th} species; $\dot{\omega}$ [kg/m²s] is the evaporation rate. $\mathbf{n}$ is the normal vector of the boundary. Eqs. (5)–(7) indicate that the net flux of aluminum is equal to the rate of evaporation of aluminum. The diffusion of aluminum is balanced by the bulk flow at the surface. For other species, the net flux at the surface is zero, i.e., there is no accumulation. The bulk velocity from the droplet phase is determined as

$$(\mathbf{u})_s = \dot{\omega} \cdot \left(\frac{1}{\rho} \right)_s \cdot \mathbf{n} = \frac{-(D_F \nabla Y_F \cdot \mathbf{n})_s}{1 - (Y_F)_s}. \quad (8)$$

The mass fraction of aluminum at the boundary, $(Y_F)_s$, is calculated based on Dalton’s law:

$$(Y_F)_s = \frac{P_{F,sat} M_F}{(P_{atm} - P_{F,sat}) M_G + P_{F,sat} M_F}, \quad (9)$$

where $P_{F,sat}$ is the saturated vapor pressure of aluminum taken from Ref. [19]. M_F and M_G are the molecular weights of aluminum and the ambient gas mixture, respectively.

The conservation of energy at the boundary in general is given as

$$(-\lambda \nabla T \cdot \mathbf{n})_s + Q_r = h_v \dot{\omega} + Q_s, \quad (10)$$

where λ and h_v are the gas phase thermal conductivity and heat of evaporation, respectively. The first term on the l.h.s in the equation denotes the conductive heat flux transferred to the droplet surface from the surrounding hot gas, the second term on the l.h.s denotes the radiative heat flux from the hot gas and alumina smoke in flame

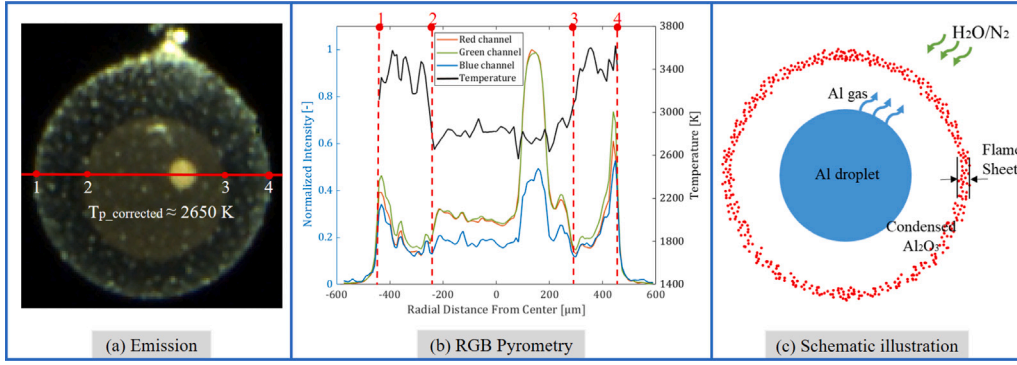


Fig. 1. Experimental measurement for a burning aluminum droplet by (a) high-speed color imaging and (b) RGB pyrometry [14], together with (c) a schematic illustration of the droplet and the enveloping flame. The droplet radius is 275 μm .

zone to the droplet surface, and the r.h.s terms denote the evaporation heat flux and heat release rate on the droplet surface, respectively. As stated previously, Q_s is assumed to be zero without considering the potential heterogeneous reaction or deposition on the droplet surface. As for Q_r , information regarding the number density and volume fraction of the alumina smoke are insufficient, resulting in difficulty and uncertainty in accurately modeling of Q_r . Considering the availability of experimentally measured surface temperature, the fixed temperature boundary condition instead of Eq. (10) was adopted in the model. However, Eq. (10) will be used to analyze the Q_r and Q_s terms later in the Discussion Sections 4.2 and 4.3.

2.3. Numerical schemes

OpenFOAM version 7 [20] is used to solve the gas phase flow and reaction. The gas phase continuity, momentum, species, and sensible enthalpy transport equations are solved using a finite volume method with a second-order accuracy scheme, e.g., the diffusion and convection terms are discretized by a filtered-linear scheme. At the same time, the time integration is based on an implicit backward scheme. The gas phase reaction is modeled using a finite rate chemistry model. The chemical reaction rates are integrated using the Seulex solver and a detailed mechanism with 21 species and 45 reactions, which is generated by extracting Al/O₂/H₂O gaseous sub-mechanism from Glorian et al. [12]. Sutherland transport model is applied, and the coefficients of each species are calculated using the third-order logarithm polynomial fitting of the species transport profile from the mechanism. The product from the gas phase reaction is observed in experiments to be nano-size liquid alumina smoke. In the mechanism, this process is described by an irreversible, fast, and barrierless reaction that generates “gaseous” Al₂O₃(l). Those “gaseous” Al₂O₃(l) molecules feature thermodynamic properties of the liquid Al₂O₃; the heat change due to condensation is thus considered. The motion of Al₂O₃ molecules is governed by the gaseous species transport equation (Eq. (2)) since the Stokes number of the nano-size liquid alumina particles is low.

3. Experimental setup

Stable generation of aluminum droplets of similar sizes under a given size of an aluminum wire is realized by positioning an aluminum wire orthogonal to the centerline of a modified McKenna burner. More details of the design can refer to the previous work of Wu et al. [14,21] and Ruan et al. [22]. In the experiment, the burner has a central H₂/O₂ diffusion flame to heat the wire to the melting point of the alumina shell (~ 2327 K [15]). Once the shell melts, the wire breaks and the aluminum droplet is generated from the wire, which then moves slowly upward with the flow. In the surroundings, a premixed H₂/O₂/N₂ flat flame provides the oxidizing environment for droplet burning.

The experimental case is the same as the C1 case in the previous work [14,21], in which three droplets with radii of 275, 120, and 55 μm are investigated. Fig. 1 shows the experimental measurement from the emission signal and temperature of an aluminum droplet with a radius of 275 μm burning in a steady state. The central droplet and the enveloping flame are visualized with high resolution. Based on Fig. 1(a), a schematic illustration of the burning of the droplet is demonstrated in Fig. 1(c). Due to the surface tension, the aluminum droplet is spherical. Aluminum continuously evaporates from the surface and reacts with the oxidizer near the droplet, forming a layer of condensed Al₂O₃(l). Important parameters to characterize the droplet evaporation and burning include flame temperature (T_f), flame stand-off ratio (STR), and Stefan flow velocity (v_{st}). With the RGB pyrometry, the droplet surface temperature and flame temperature are measured to be ~ 2650 K and ~ 3400 K for all the droplet sizes. With the resolved droplet image, the droplet surface and its surrounding flame zone can be anchored to obtain STR (the ratio of the position of the peak temperature over the position of the droplet surface, i.e., R_f/R). Moreover, the droplet size evolution [m^2/s] can be obtained from the resolved images by fitting to a D²-law [23]. Then the evaporation rate $\dot{\omega}$ can be calculated as per $\dot{\omega} = d(\rho_l \cdot \pi D^3/6)/dt$, from which v_{st} can be further calculated. The experimental results are shown in Table 1 together with the simulation results.

Detailed numerical simulation of the H₂/O₂/N₂ flame shows that during the steady state burning stage of the droplet, the surrounding temperature and oxidizer concentration can be considered constant at ~ 2600 K with the composition of 55%H₂O/45%N₂ by mole. This information will be used as the setup condition in the simulation. A constant droplet surface temperature of 2650 K will also be adopted in the simulation. Target parameters (T_f , STR, and v_{st}) will be computed and compared with experimental data. T_f is taken as the peak value in the temperature profile. STR is taken as the ratio of the position of the peak temperature over the position of the droplet surface (R_f/R). v_{st} is the normal component of the bulk flow velocity ($\mathbf{u}_s$) in Eq. (8).

4. Results and discussion

4.1. Grid geometry and validation

In correspondence to the experimental measurement, Fig. 2 shows the computational domain used in the simulation. Different from the 2D (r, θ) cylindrical domain adopted by many other authors [12,13,17], this work employs a spherical symmetrical pseudo-1D computational domain to increase the computation efficiency. The droplet surface is treated as the boundary of the domain. Initially, the gas phase region is filled with oxidizer. Gaseous aluminum flows out of the Fuel boundary with a velocity of ($\mathbf{u}_s$), then mixes and reacts with the oxidizer. The size of the computational domain is determined so that there is enough

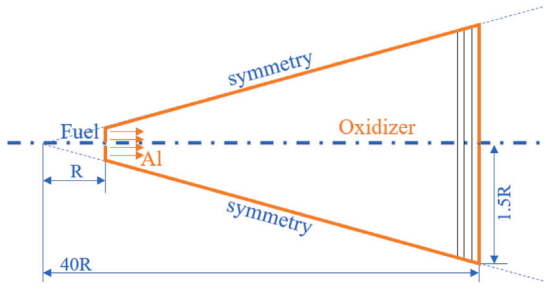


Fig. 2. Illustration of the pseudo-1D simulation domain.

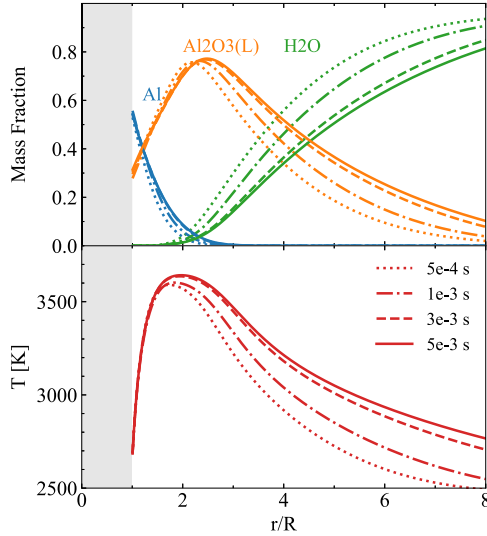


Fig. 3. Temporal evolution of the temperature and main species profiles in simulation. The radial distance is normalized by the initial droplet radius. Droplet radius: 275 μm . Oxidizing condition: 2600 K in pure H_2O .

oxidizer to sustain the reaction until the computation ends. As a steady-state solution is pursued, the end time is set so that temperature and species profile variation is smaller than 2%, as seen in Fig. 3. The mesh size is set to 2 μm , which is $\sim 0.036R$ for the smallest droplet. A grid-independent numerical solution is obtained by adopting different mesh sizes, from 1.5 to 5.5 μm , and simulation results from these meshes overlap for temperature and main species distributions within the investigated region, i.e., the region between the droplet surface and the reaction zone. The detailed information on temperature and species distribution in this region can be accurately resolved with the 2 μm mesh size, which is used in the following discussion.

4.2. Temperature and species distribution

Fig. 4 presents typical steady-state simulation results of temperature, Stefan flow velocity, and species distribution in the vicinity of a burning aluminum droplet. The simulation is done for a droplet radius of 275 μm with the oxidizer condition of pure H_2O at 2600 K. The boundary temperature is set constant at 2650 K. Temperature profile starts with a sharp increase from the *Fuel* boundary to peak value, denoted herein as flame temperature T_f , and then gradually decreases due to heat dissipation to the surrounding. The determination of T_f , v_{st} , and STR from the temperature profile is shown in the figure. The velocity is the highest near the boundary and then gradually decreases due to expansion. Species with a maximum mass fraction larger than 3% are plotted in Fig. 4(b). The species distribution profile shows a classical diffusion flame structure [24].

On the fuel side, Al has the highest mass fraction and gradually decreases outwards, while an opposite trend is observed for steam. The main reaction product calculated from the current mechanism is $\text{Al}_2\text{O}_3(l)$, which has significant concentration over the region and peaks near the flame temperature. Apart from that, AlOAl also has a non-negligible contribution within $3R$ from the surface. OH begins to form after the peak of AlOAl and diffuses quite fast to the surrounding gas. It is interesting to note that the concentration of AlO is negligible in the steam case, although it is often interpreted as a marker for the flame position when oxygen is the oxidizer [6]. Other intermediate species include some Al/O/H sub-oxides such as AlH , AlOH , and AlOAlO . However, their mass fractions are much lower than those plotted in the figure. Meanwhile, due to the species diffusion, products can reach the droplet surface, opening up some possibility for surface reaction or deposition.

Furthermore, although Eq. (10) is not implemented in the current model, the first term on the l.h.s and the first term on the r.h.s of the equation can be used to calculate heat fluxes through the *Fuel* boundary. Based on the results from Fig. 4, the evaporation rate $\dot{m}$ is calculated to be $\sim 0.726 \text{ kg/m}^2\text{s}$ from Eq. (8). The specific evaporation heat of Al at 2650 K is $\sim 10.94 \text{ MJ/kg}$; thus, the evaporation heat flux, given by the first term on the r.h.s of Eq. (10), is approximately 7.94 MW/m^2 . On the other hand, the conductive heat flux transported from the reaction zone to the boundary is calculated to be $\sim 2.55 \text{ MW/m}^2$ as per the first term on the l.h.s of Eq. (10). It is somewhat surprising to find that the conductive heat only accounts for $\sim 32\%$ of the heat needed to sustain Al evaporation. Note that the calculated evaporation velocity oscillates between 5 and 10 m/s, which might slightly influence the evaporation heat flux. However, in any case, it is more than twice the conductive heat flux from the flame zone. Meanwhile, the calculation for the other two droplet sizes shows similar ratios of the heat from the reaction zone to that needed for evaporation. This demonstrates that the inclusion of other heat source terms, i.e., Q_r and/or Q_s , are indispensable. The evaporation heat cannot be balanced solely by heat conduction from the flame zone. This imbalance will be further investigated in the following sub-section.

4.3. Dependence of droplet diameter and oxidizer concentration

Table 1 compares the predicted flame parameters T_f , STR, and v_{st} against corresponding experimental data. Three droplets with radii of 275, 120, and 55 μm are investigated.

Experimental results find that the surface temperature and flame temperature do not vary much for different droplet sizes under the current ambient hot steam condition. The measured STR shows a moderate decrease with increasing droplet size. The droplet burning rate, which is related to the Stefan flow velocity (v_{st}), decreases with increasing droplet size.

Meanwhile, results from the simulation show that all the predicted values are of the same order of magnitude as those of the experiments. The flame temperatures deviate less than 5% of the experimental value. The STR and v_{st} of smaller droplets (55 μm radius) agree quite well with experiments, yet larger STR and smaller v_{st} are obtained for larger droplets (radii of 120 and 275 μm). It should be pointed out that the current model simulates the burning of a single Al droplet in a quiescent environment, while in experiments, the droplet is moving together with the surrounding flow. Smaller droplets can follow the flow better. For large droplets, both the diameter and slip velocity are larger, which result in a larger Reynolds number, and thus, a larger Nusselt number to enhance evaporation and droplet burning [23].

It is also interesting to note the dependence of flame parameters on droplet sizes. Fig. 5 plots the detailed temperature and species profiles for the three droplet sizes. Although the steady-state conditions are modeled separately for three droplet sizes, combining them, the evolution of the droplet burning, i.e., different statuses as the droplet

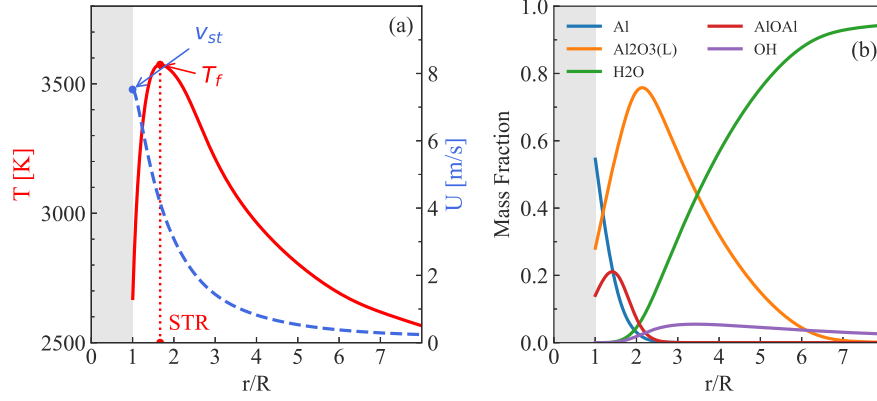


Fig. 4. Numerically predicted temperature, Stefan flow velocity, and main species distribution in the vicinity of an aluminum droplet surface. Droplet radius: 275 μm . Oxidizing condition: 2600 K in pure H_2O .

Table 1

Experimental and computational cases, and key parameters from the experiments and numerical simulation. Oxidizer condition: 55% H_2O /45% N_2 by mole at 2600 K.

Radius [μm]	Simulation			Experiment			
	T_f [K]	STR	v_{st} [m/s]	T_p [K]	T_f [K]	STR	v_{st} [m/s]
275	3496	2.12	6			1.75	15
120	3545	2.3	10	2650	3400	1.83	20
55	3555	2.11	24			2	26

evaporates and shrinks, can be revealed. The predicted flame temperatures do not vary much under three droplet sizes, with a slight increase within 60 K, which agrees with experimental data. The predicted STRs also do not vary much with a slight increase and then decrease as the droplet size increases, while the experiment result shows a slightly decreasing trend of STR as droplet size increases. It is worth noting that in experiments, larger droplets feature a larger slip velocity. This exerts a “stretching” effect on the droplet and its surrounding flame, leading to non-concentricity. Smaller droplets are observed to follow the flow better and are closer to symmetric combustion. Thus, STR is more accurately determined for the smaller droplets. On the contrary, symmetric combustion with no slip velocity is considered in the simulation. In this sense, the case of smaller droplets is closer to the simulation setup, which is also evidenced by the closer alignment of the predicted STR and the measured STR for the 55 μm radius case. The Stefan flow velocity decreases as droplet size increases, with the predicted values decreasing much faster than those results from the experiment. As per Eq. (8), if the boundary temperature were constant, the Al partial pressure at the boundary would be constant. Considering that the species mass fraction at the boundary is almost the same, ρ and D will also be almost the same. Then, Stefan flow velocity is almost proportional to the gradient of Al at the boundary. From Fig. 5, the Al profiles overlap near the boundary along the normalized radius, which indicates that the actual gradient is almost inversely proportional to the droplet radius. Then, it is reasonable that the simulated Stefan flow velocity increases as droplet size decreases.

Heat flux analysis is also performed for the above-simulated cases. Results show that approximately 27%, 22%, and 18% of evaporation heat is provided by the heat conduction from the flame for droplet radii of 275, 120, and 55 μm , respectively. If we look at the 55 μm radius case in which all the predicted flame parameters match well with experiment data, heat conduction from the flame still only accounts for 18% of the evaporation heat needed. This imbalance of evaporation heat and conduction heat also changes with varying droplet surface temperature. Table 2 lists the simulation results with boundary temperatures between 2350 and 2650 K. As boundary temperature decreases, the flame temperature decreases accordingly, together with

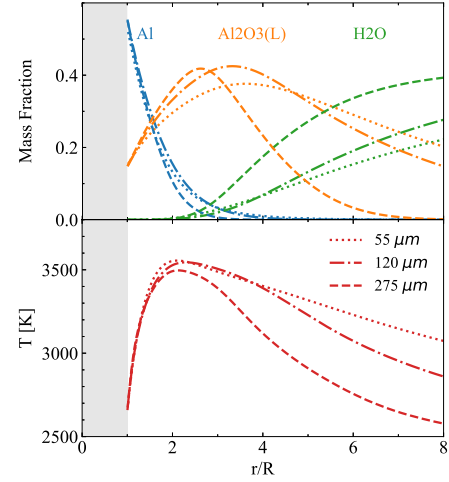


Fig. 5. Predicted temperature and species profiles along the normalized radius for different droplet sizes ($R=275, 120, 55 \mu\text{m}$) under the experimental condition.

Table 2

Predicted flame parameters for varying boundary temperatures under the experiment condition (55% H_2O /45% N_2 by mole at 2600 K). The droplet radius is 55 μm .

Radius [μm]	T_p [K]	T_f [K]	STR	v_{st} [m/s]	Q_f/Q_{evap} [%]
55	2650	3555	2.11	24	18
	2600	3500	1.94	20	26
	2500	3255	1.86	14	45
	2400	3055	2.68	10	61
	2350	2994	3.06	8	69
55 ^a	2650	3550	2.08	23	18

^a Results with the inclusion of a NO_x sub-mechanism.

Stefan flow velocity. STR first decreases and then increases as boundary temperature decreases. There is a significant increase in the proportion of the conduction heat over the evaporation heat from 18% at 2650 K to 69% at 2350 K. Further lowering the temperature will reach the solidification/melting point of $Al_2O_3(L)$, which is not physical in that the burning of Al droplet is generally considered to be activated after the melting of the Al_2O_3 coating layer [10,17]. This result reveals that the heat imbalance becomes more significant at a higher surface temperature, suggesting the increasing importance of other heat sources that should be further quantified.

A parameter study of the influence of oxidizer (H_2O) fraction was also conducted. Fig. 6 shows the predicted flame parameters if the N_2 in C1 case is removed, i.e., pure H_2O as the oxidizer. If the temperature

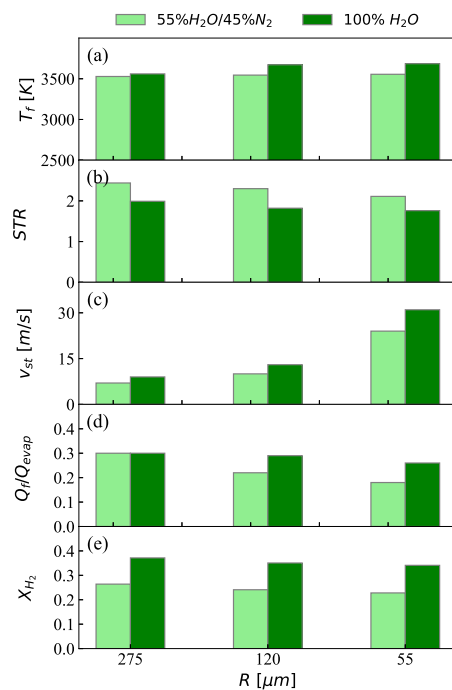


Fig. 6. Effects of H₂O fractions in the oxidizer on predicted (a) flame temperature, (b) stand-off ratio, (c) Stefan flow velocity, (d) heat ratio, and (e) generated H₂ amount (maximum mole fraction). Droplet radius: 275, 120, and 55 μm. Boundary temperature set at 2650 K.

is fixed at 2650 K, an increase of the H₂O fraction in the oxidizer leads to a slight increase in the flame temperature (~100 K), a smaller STR, and a higher Stefan flow velocity. The higher Stefan flow velocity indicates a higher evaporation rate, which is as expected. The ratio of conduction heat over evaporation heat is higher in pure H₂O compared with the N₂-diluted case, as droplet size increases. If a comparable ratio is to be obtained in the steam case as that in the N₂-diluted case, then the boundary temperature in the pure steam case would be set higher. Maximum H₂ mole fraction is not affected by droplet sizes, while a higher concentration of H₂O leads to a higher maximum H₂ mole fraction. With the mole fraction of H₂O increasing from 55% to 100%, which is ~1.8, the increase of the maximum H₂ mole fraction, however, is yet less than 1.5. The effects of the increase in H₂O fraction on H₂ generation are not that evident.

NOx emission is an important issue to be considered in metal combustion when N₂ exists in the oxidizer since the flame temperature is very high, as discussed earlier. To investigate this, a NOx mechanism, which was extracted from a reduced NH₃/H₂ mechanism [25] with 28 species and 213 reactions, was integrated into the current mechanism to calculate the NOx formation in the C1 case for the 55 μm radius case. As shown in the first and the last row of Table 2, the inclusion of the NOx mechanism leads to a 5 K (0.14%) decrease in the flame temperature, 1.4% decrease in STR, 4% decrease in Stefan flow velocity, 4% increase in maximum H₂ mole fraction (not shown in the table), and negligible impact on Q_f/Q_{evap} , which demonstrates the weak impact of NOx chemistry on the overall flame structure. Regarding the NOx emission, the maximum mole fractions of NO₂ and N₂O are below 0.5 ppm. In contrast, the maximum mole fraction of NO can reach up to ~2050 ppm, which is significantly higher than typical Dry-Low Emission (DLE) gas turbine combustion of 25 ppm [26]. In the future Al/steam combustion design, N₂ fraction in the oxidizer should be lowered to avoid high NOx emission.

5. Conclusions

This work presents a detailed numerical investigation of the flame structure and burning rate of a micron-sized aluminum droplet in hot steam-dominated environments in comparison with a new set of experimental data. The numerical model incorporates detailed mass and heat transfer between the gas phase and droplet phase, along with a detailed chemical kinetic mechanism for aluminum vapor reaction. A pseudo-1D spherical symmetric computational domain was employed to better represent the spherical droplet surface as the boundary.

Experimental results show that the surface temperature and flame temperature do not vary much for different droplet sizes investigated. However, the stand-off ratio decreases when the droplet size decreases.

The simulated flame exhibits a classical diffusion flame structure close to the droplet, whereas, near the boundary, there is a non-negligible amount of AlOAl apart from the main product Al₂O₃(l). The simulation of the experimental case with the current model shows agreeable consistency in flame temperature for all the droplet sizes, and the flame stand-off ratio and the Stefan flow velocity for small droplet sizes (radius of 55 μm).

Heat flux analysis shows that conduction heat from the flame front accounts for less than 30% of the heat needed in Al evaporation, and the imbalance becomes more significant as droplet surface temperature increases. The increase of H₂O fraction in the oxidizer does not seem to significantly impact the predicted flame parameters, especially for large droplet sizes (radius of 275 μm), indicating that H₂O may be a weak oxidizer. Furthermore, simulation results show that NOx emission should be considered in the investigated H₂O/N₂ environments.

Overall, the current model serves as a promising platform to investigate the effects of different oxidizing conditions (oxidizer composition and temperature) on the flame parameters and to examine different chemical mechanisms. Further improvement of the model will benefit from the quantification of the energy conservation using combined experimental and computational methods so as to fully describe the involved physical and chemical processes.

Novelty and significance statement

The innovation of this research lies in a joint experimental and numerical study of detailed flame structure, flame temperature, droplet temperature, and flame stabilization position of micron-sized aluminum droplets. The results are valuable for a quantitative description of the aluminum/steam reaction for the production of hydrogen, which will impact the development of technologies for the generation of carbon-free fuels addressing the future green transition of energy and power industries.

A detailed numerical model was developed and tested to resolve the boundary layer of steadily burning aluminum droplets. Detailed temperature, velocity, and species profiles can be simulated and validated against a set of new experimental data, including flame temperature, flame stand-off ratio, and Stefan flow velocity. The results from net heat flux analysis are another novelty that opens up further exploration of other heat sources or some defect in the current mechanism.

CRediT authorship contribution statement

Yue Qiu: Model development, Results analysis, Writing – original draft. **Sheng Feng:** Results analysis, Software, Review. **Zhiyong Wu:** Experiments, Data analysis, Review. **Shijie Xu:** Software, Review. **Can Ruan:** Experiments, Data analysis, Review. **Xue-Song Bai:** Conceptualization, Supervision, Results and Discussion. **Elna J.K. Nilsson:** Project administration, Supervision, Introduction. **Marcus Aldén:** Project leader, Supervision, Review. **Zhongshan Li:** Experiment, Supervision, Review.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Knut and Alice Wallenberg Foundation, Sweden within the COCALD project. The computations were enabled by resources provided by the National Academic Infrastructure for Supercomputing in Sweden (NAISS) and the Swedish National Infrastructure for Computing (SNIC) at NSC, HPC2N, and PDC, partially funded by the Swedish Research Council through grant agreements no. 2022-06725 and no. 2018-05973, and for awarding access to the LUMI supercomputer, owned by the EuroHPC Joint Undertaking, hosted by CSC (Finland) and the LUMI consortium.

References

- [1] K. Trowell, S. Goroshin, D. Frost, J. Bergthorson, Aluminum and its role as a recyclable, sustainable carrier of renewable energy, *Appl. Energ.* 275 (2020) 115112.
- [2] F. Halter, S. Jeanjean, C. Chauveau, Y. Berro, M. Balat-Pichelin, J.F. Brilhac, A. Andrieu, C. Schonnenbeck, G. Leyssens, C. Dumand, Recyclable metal fuels as future zero-carbon energy carrier, *Appl. Energy Combust. Sci.* 13 (2023) 100100.
- [3] R. Friedman, A. Maček, Ignition and combustion of aluminium particles in hot ambient gases, *Combust. Flame* 6 (1962) 9–19.
- [4] A. Davis, Solid propellants: The combustion of particles of metal ingredients, *Combust. Flame* 7 (1963) 359–367.
- [5] M. Gurevich, G. Ozerova, A. Stepanov, Heterogeneous ignition of an aluminum particle in oxygen and water vapor, *Combust. Explos. Shock Waves* 6 (3) (1970) 291–297.
- [6] P. Bucher, R. Yetter, F. Dryer, T. Parr, D. Hanson-Parr, E. Viceni, Flames structure measurement of single, isolated aluminum particles burning in air, *Proc. Combust. Inst.* 26 (1996) 1899–1908.
- [7] P. Bucher, R. Yetter, F. Dryer, E. Vicenzi, T. Parr, D. Hanson-Parr, Condensed-phase species distributions about Al particles reacting in various oxidizers, *Combust. Flame* 117 (1–2) (1999) 351–361.
- [8] P. Bucher, L. Ernst, F.L. Dryer, R.A. Yetter, T.P. Parr, D.M. Hanson-Parr, Detailed studies on the flame structure of aluminum particle combustion, in: *Solid Propellant Chemistry, Combustion, and Motor Interior Ballistics*, in: *Prog. Aerosp. Sci.*, American Institute of Aeronautics and Astronautics, ISBN: 978-1-56347-442-2, 2000, pp. 689–722.
- [9] F. Halter, V. Glasziou, M. Di Lorenzo, S. Gallier, C. Chauveau, Peculiarities of aluminum particle combustion in steam, *Proc. Combust. Inst.* 39 (3) (2022) 3605–3614.
- [10] M. Beckstead, Correlating aluminum burning times, *Combust. Explos. Shock Waves* 41 (5) (2005) 533–546.
- [11] J. Zhang, O.T. Stein, A. Shamooni, X. Zhixun, L. Zhenbing, M. Likun, F. Yunchao, A. Kronenburg, Detailed modeling of aluminum particle combustion—From single particles to cloud combustion in Bunsen flames, *Chinese Journal of Aeronautics* 35 (5) (2022) 319–332.
- [12] J. Glorian, S. Gallier, L. Catoire, On the role of heterogeneous reactions in aluminum combustion, *Combust. Flame* 168 (2016) 378–392.
- [13] Z.J. Hu, T. Yang, Z.X. Xia, L.K. Ma, M.T. Li, Y.C. Feng, The simulation of different combustion stages of micron-sized aluminum particles, *Appl. Sci.* 11 (4) (2021) 1774.
- [14] Z. Wu, C. Ruan, M. Stiti, S. Xu, X.-S. Bai, E. Berrocal, M. Aldén, Z. Li, Spatiotemporally resolved surface temperature measurement of aluminum ignition and combustion in steam and oxygenated environments, *Combust. Flame* 264 (2024) 113446.
- [15] M.W. Chase, NIST-JANAF thermochemical tables, *J. Phys. Chem.* 9 (1998).
- [16] M. Leitner, T. Leitner, A. Schmon, K. Aziz, G. Pottlacher, Thermophysical properties of liquid aluminum, *Metall. Mater. Trans. A* 48 (2017) 3036–3045.
- [17] M. Beckstead, A summary of aluminum combustion, Report No. N00014-95-1-1338, Brigham Young Univ Provo Ut, 2004.
- [18] T. Nambu, Y. Mizobuchi, A study on droplet group evaporation modeling based on interface resolved numerical simulations of two-phase flow, *Combust. Flame* 248 (2023) 112549.
- [19] R.E. Honig, Vapor pressure data for the solid and liquid elements, *RCA Rev.* 23 (12) (1962) 567.
- [20] H.G. Weller, G. Tabor, H. Jasak, et al., A tensorial approach to computational continuum mechanics using object-oriented techniques, *Comput. Phys.* 12 (6) (1998) 620–631.
- [21] Z. Wu, C. Ruan, M. Stiti, Y. Qiu, S. Xu, N. Jüngst, E. Berrocal, M. Aldén, X.-S. Bai, Z. Li, Flame structure and evaporation rate of single aluminum droplets burning in hot steam-dominated flows, *Combust. Flame*, under review (2024).
- [22] C. Ruan, Z. Wu, M. Stiti, M. Snellman, Y. Qiu, A.A. Subash, E. Berrocal, M. Aldén, Z. Li, Combustion of micron-sized Al-Mg alloy wires in hot $H_2O/O_2/N_2$ flows, *Fuel* 357 (2024) 129719.
- [23] K.P. Brooks, M.W. Beckstead, Dynamics of aluminum combustion, *J. Propuls. Power* 11 (4) (1995) 769–780.
- [24] C.K. Law, *Combustion Physics*, Cambridge University Press, 2010.
- [25] R. Li, A.A. Konnov, G. He, F. Qin, D. Zhang, Chemical mechanism development and reduction for combustion of $NH_3/H_2/CH_4$ mixtures, *Fuel* 257 (2019) 116059.
- [26] E. Jeffs, Low emissions Rolls-Royce aeroderivative below 25 ppm NOx, *Turbomach. Int.* 34 (6) (1993).